# Problem Set 4

Zahra Khan

Due: April 10, 2026

## Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Friday April 10, 2026. No late assignments will be accepted.

## Question 1

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the `eha` library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

Table 1:

| | *Dependent variable:* |
|---|---|
| | child_surv |
| sexfemale | $-0.082^{***}$ |
| | (0.027) |
| m.age | $0.008^{***}$ |
| | (0.002) |
| Observations | 26,574 |
| $R^2$ | 0.001 |
| Max. Possible $R^2$ | 0.986 |
| Log Likelihood | $-56{,}503.480$ |
| Wald Test | $22.520^{***}$ (df = 2) |
| LR Test | $22.518^{***}$ (df = 2) |
| Score (Logrank) Test | $22.530^{***}$ (df = 2) |
| *Note:* | $^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01 |

There is a 0.08 decrease in the expected log of the hazard for female babies compared to male, holding mothers age constant. There is a 0.008 increase in the expected log of the hazard for babies of older mothers compared to younger mothers, holding sex constant.

# Question 2

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the `disasters_response` data on GitHub in which `binContribution` is the outcome for the selection equation, and `originalContributionMillionUSDLogged` is the outcome for the second equation. Use `occurrences + deathsEM + normalizedDamageEMLogged` as your input variables for both equations. Present and interpret the output.